Introduction to Artificial Intelligence and Semantic Search

Artificial Intelligence (AI) is a branch of computer science that develops systems capable of performing tasks that normally require human intelligence. These tasks include reasoning, learning, planning, problem-solving, language understanding, and decision-making.

Machine Learning (ML) is a subset of Artificial Intelligence. It enables computers to learn from data without being explicitly programmed. Machine learning algorithms identify patterns in data and use them to make predictions or decisions.

Natural Language Processing (NLP) is an important area of AI that focuses on enabling computers to understand, interpret, and generate human language. Applications of NLP include chatbots, language translation, text summarization, and question-answering systems.

Large Language Models (LLMs) such as Llama and GPT are trained on massive collections of text. These models can answer questions, summarize documents, write code, generate text, and assist users with many different tasks.

Embeddings are numerical vector representations of text. They capture the semantic meaning of words, sentences, or documents. Text with similar meanings produces similar embeddings even when different words are used.

Semantic Search is a search technique that finds information based on meaning instead of exact keyword matching. It uses embeddings and similarity metrics to retrieve the most relevant documents.

Cosine similarity is the most common similarity metric used in semantic search. It measures the angle between two embedding vectors. A higher cosine similarity indicates that the vectors represent similar meanings.

Vector databases such as ChromaDB store embeddings efficiently. They support fast similarity searches and metadata filtering, making them suitable for Retrieval-Augmented Generation (RAG) applications.

Retrieval-Augmented Generation (RAG) combines document retrieval with a Large Language Model. Instead of answering from memory alone, the system retrieves relevant document chunks from a vector database and provides them to the LLM as context before generating an answer.

Ollama allows developers to run open-source language models locally on their computers. This provides better privacy, offline access, and lower latency compared to cloud-based services.

Modern AI applications such as document search, enterprise knowledge assistants, customer support chatbots, and intelligent search engines rely on embeddings, vector

databases, semantic search, and Retrieval-Augmented Generation to deliver accurate and context-aware responses.